

Model A · Diabetic Retinopathy Grading · Full Fine-tune

All RETFound-DINOv2 layers updated · Layer-wise LR decay · 50 epochs · 4-class (R0/R1/R2/R3A)

Upgrade from Linear Probe → Full Fine-tune

LP kept the ~307 M backbone FROZEN — only a single linear head was trained, measuring how much structure was already encoded.

FT UNFREEZES all layers with layer-wise LR decay (factor 0.65): deepest blocks get full LR; shallowest get ~0.0002× LR. This allows non-linear class boundaries that a linear head cannot form.

0.922

Test AUROC (LP 0.830)

0.614

Macro Sensitivity (LP 0.578)

0.921

Macro Specificity (LP 0.868)

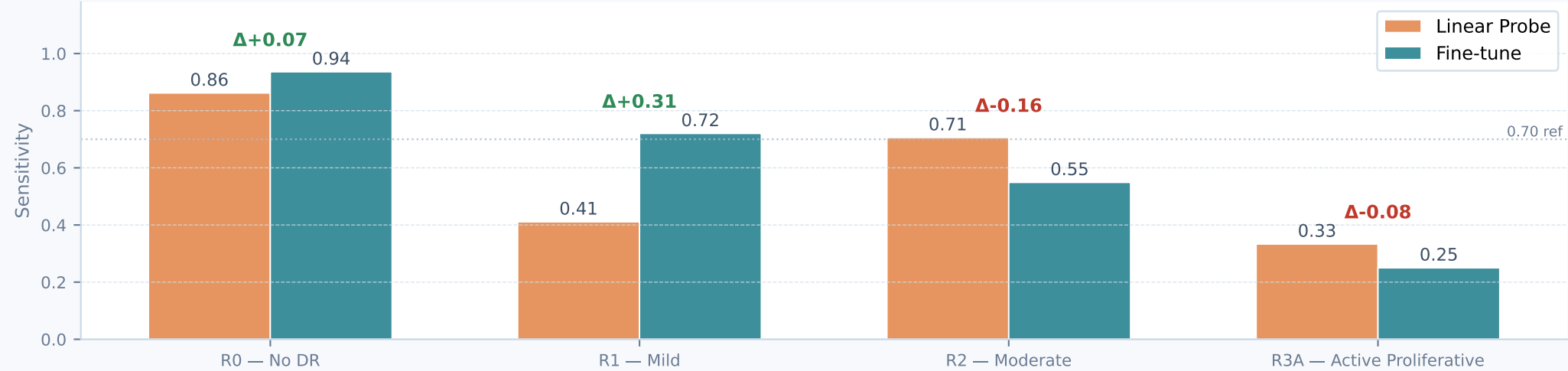
FT Settings (differences from Linear Probe)

Backbone	FULLY UNFROZEN (LP: frozen — no gradient)
LR scheme	Layer-wise decay ×0.65 per depth level
Batch size	24 (LP: 64 — reduced to fit full backprop GP)
Best epoch	7 (LP: 38 — early convergence with unfroze)
Epochs trained	50 · Best ckpt saved by val AUROC
Loss / weights	CrossEntropyLoss · R0=1.00 R1=1.79 R2=9.
Test eval	Best ckpt reloaded → single pass on held-out

LP vs FT · Test Set Metrics

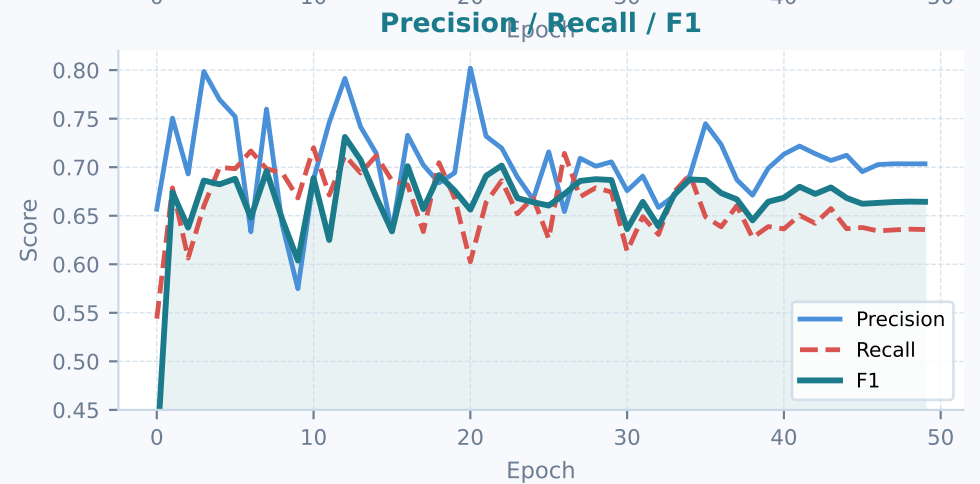
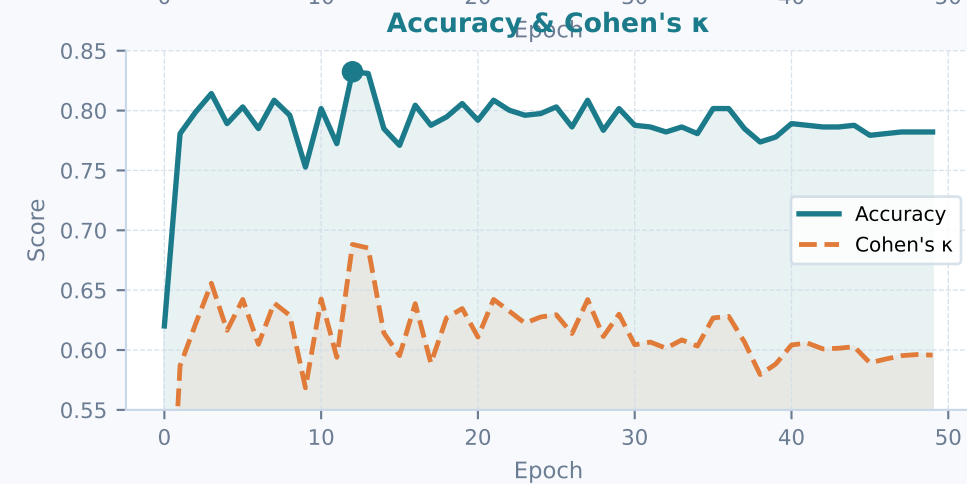
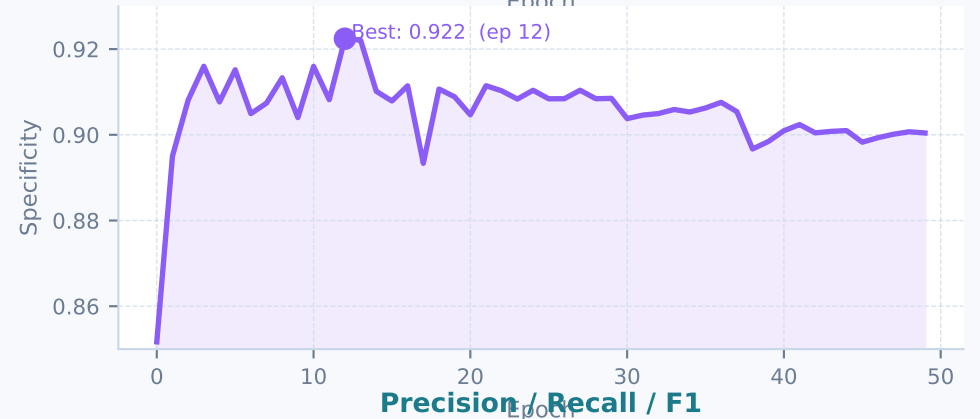
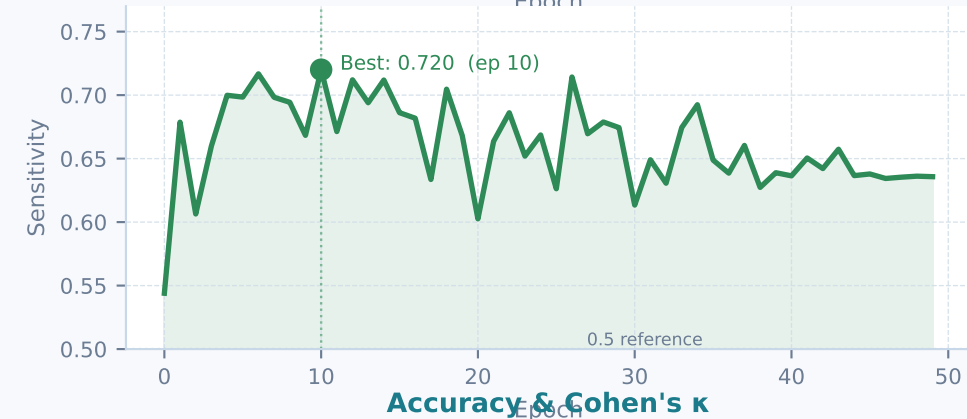
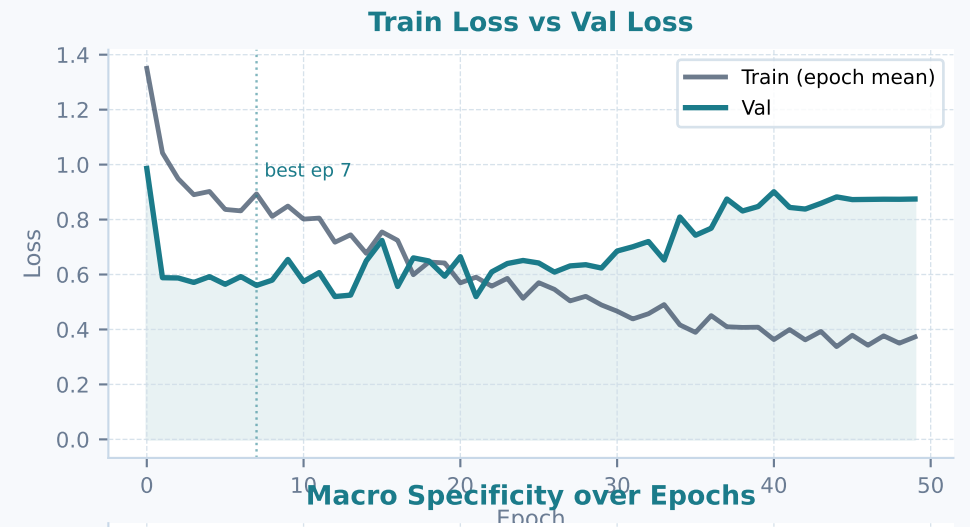
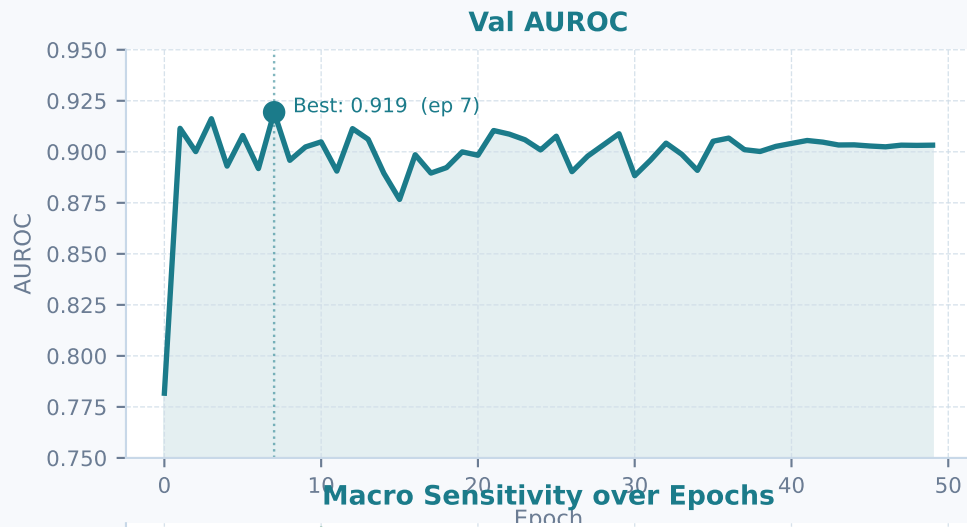
Metric	Linear Probe	Fine-tune	Δ
AUROC	0.830	0.922	+0.093
Accuracy	68.1%	81.2%	+13.1%
F1 (macro)	0.533	0.633	+0.100
Kappa	0.446	0.663	+0.217
Macro Sensitivity	0.578	0.614	+0.036
Macro Specificity	0.868	0.921	+0.053

Per-class Sensitivity: Linear Probe vs Fine-tune (test set)



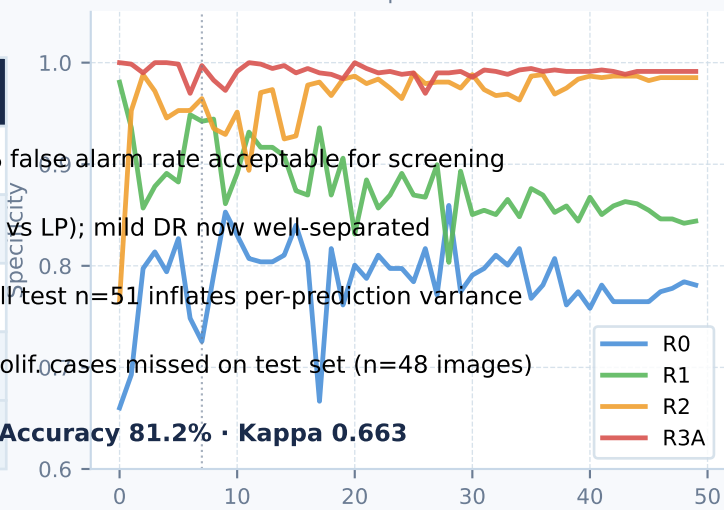
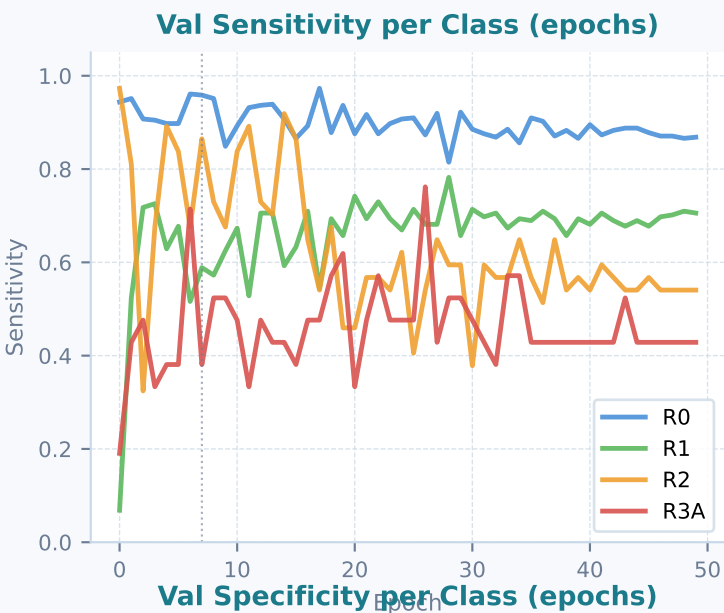
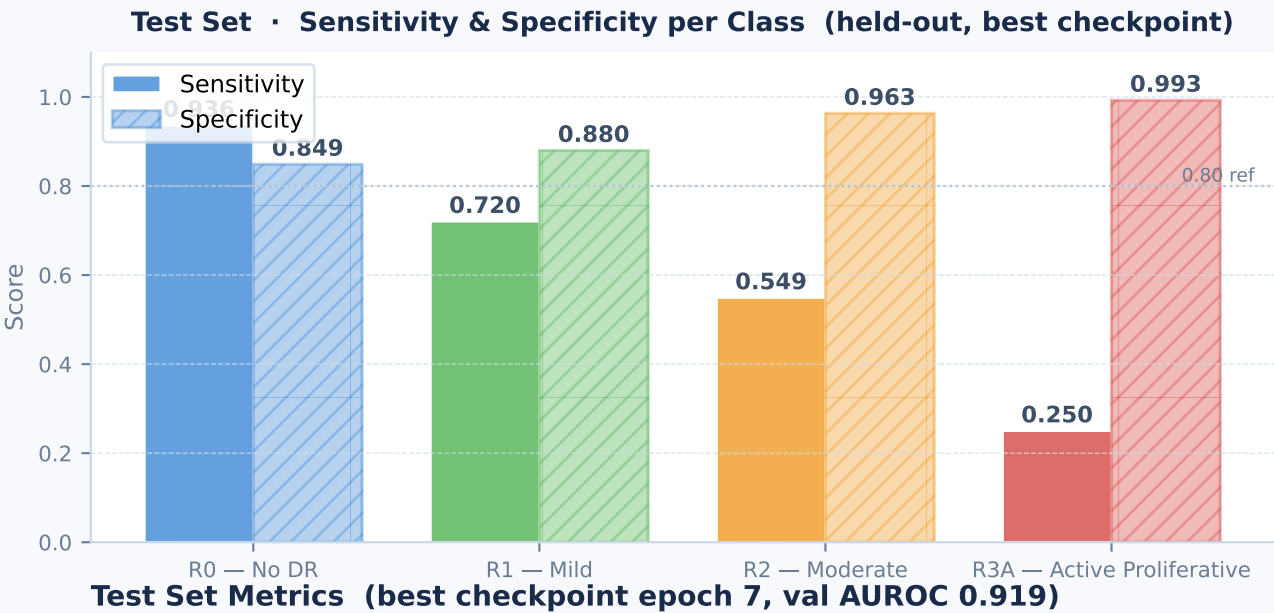
Training Dynamics — All Validation Metrics over 50 Epochs

Best checkpoint at epoch 7 (val AUROC 0.9194) · All backbone layers updated each step



Per-class Sensitivity & Specificity — Full Fine-tune (TEST SET)

Sensitivity = TP/(TP+FN) · Specificity = TN/(TN+FP) · OvR decomposition at best-AUROC checkpoint



Class	Sensitivity	Specificity	Clinical Note
R0 — No DR	0.936	0.849	Excellent recall; ~15% false alarm rate acceptable for screening
R1 — Mild	0.720	0.880	Major FT gain (+0.309 vs LP); mild DR now well-separated
R2 — Moderate	0.549	0.963	Below LP (0.706); small test n=51 inflates per-prediction variance
R3A — Active Prolif.	0.250	0.993	Critical gap: 75% of proliferative cases missed on test set (n=48 images)
Macro average	0.614	0.921	Test AUROC 0.922 · Accuracy 81.2% · Kappa 0.663

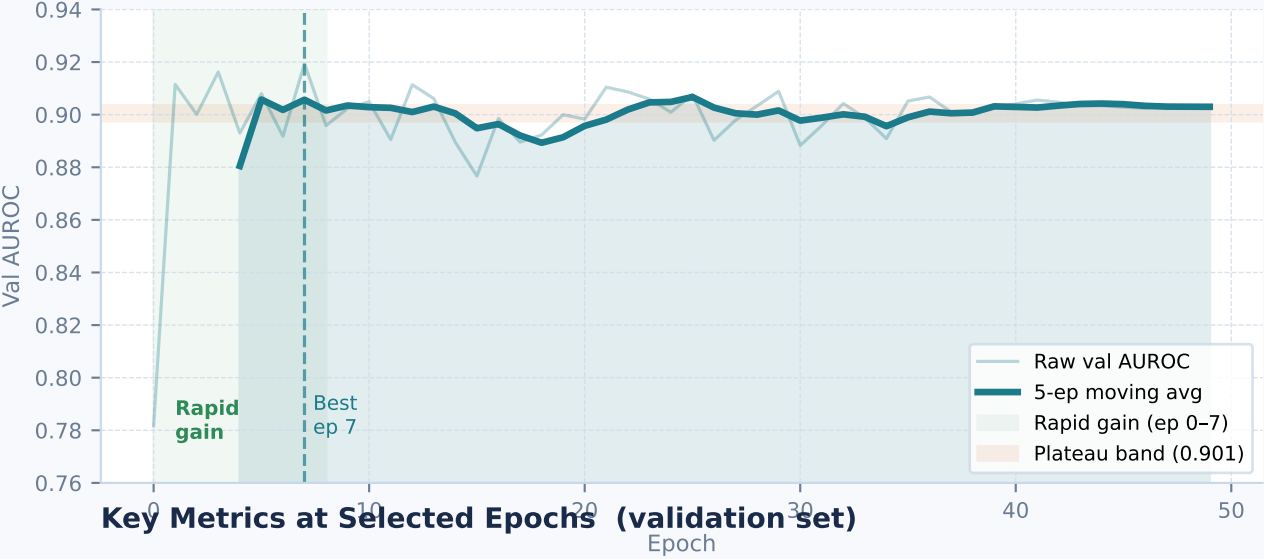
Clinical Interpretation of Fine-tuned Per-class Performance

- **R3A sensitivity (0.250):** 75% of proliferative cases go undetected — the most dangerous failure. Low test n (48) means each missed prediction moves sensitivity by 0.021; interpret with caution.
- **R1 recovery (+0.309 vs LP):** Biggest win of fine-tuning: mild DR jumped from 0.411 to 0.720. Backbone adaptation enables non-linear class boundaries the frozen linear head could not form.
- **R2 regression (−0.157 vs LP):** Moderate DR sensitivity dropped from 0.706 to 0.549. Likely the model re-allocates R2 feature space to better separate R1, trading moderate recall for mild recall.

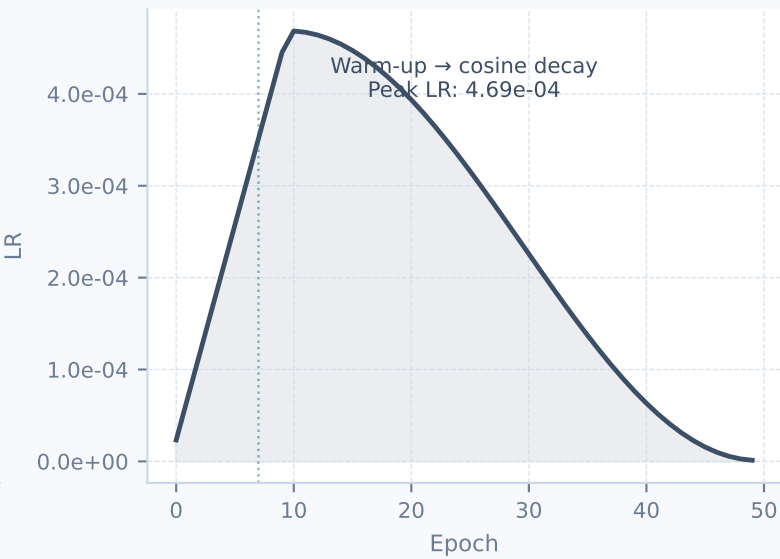
Convergence Analysis · LR Schedule · Milestone Epochs

Best ckpt at epoch 7 · Plateau ~0.901 from epoch 10 onwards · Val loss divergence = overfitting signal

AUROC Convergence (raw + 5-epoch moving average)



Learning Rate Schedule



Epoch	AUROC	Accuracy	Kappa	Macro Sensitivity	Macro Specificity	Train Loss	Val Loss
Ep 0	0.7817	0.6201	0.3456	0.5440	0.8517	1.3504	0.9860
Ep 3	0.9162	0.8142	0.6558	0.6599	0.9160	0.8904	0.5711
Ep 7 ★	0.9194	0.8087	0.6392	0.6983	0.9074	0.8933	0.5604
Ep 12	0.9114	0.8324	0.6882	0.7120	0.9224	0.7175	0.5198
Ep 24	0.9009	0.7975	0.6276	0.6686	0.9104	0.5134	0.6510
Ep 49	0.9033	0.7821	0.5957	0.6358	0.9004	0.3730	0.8748

Key Observations

- AUROC jumped 0.782 → 0.919 in just 7 epochs — pretrained backbone features activate rapidly when all layers receive gradients.
- Val AUROC peaked at epoch 7 then oscillated ~0.901 while val loss climbed 0.56 → 0.87: a clear overfitting signal. Early stopping (patience ≈ 10 on val AUROC) would save ~42 epochs of compute.
- Train loss continued declining to 0.337 by epoch 44, confirming the model kept learning on training data after val AUROC plateaued— the gap between train and val loss widens steadily past epoch 10.
- LR warm-up ramps over the first 10 epochs, peaking at 4.69e-04, then cosine decays to near-zero— this schedule matches the AUROC curve's rapid early rise followed by diminishing returns.
- Sensitivity and specificity at the best checkpoint (epoch 7) reflect the sharpest moment of generalisation; later epochs improve train metrics but not val/test performance.
- Recommended next step: add early stopping to the fine-tune script to avoid unnecessary epochs; then proceed to Model B (maculopathy binary task).